Supplementary Material
Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

Products. Every number in this document is read from the same committed products as the main paper: results/main/replay_summary.csv, results/main/fixed_references.csv, results/main/tau_feasible.csv, results/main/true_e2e_ap.csv, results/main/action_distribution.csv, results/main/feature_importance_frozen.csv, results/main/step4_collaboration_harm.csv, results/sensitivity/feature_ablation.csv, results/sensitivity/difficulty_frozen.csv, results/sensitivity/collaborator_scale.csv, results/sensitivity/delivery_semantics_bracket.csv, results/sensitivity/robustness_frozen.csv, results/sensitivity/r23_scene_bootstrap.csv, results/sensitivity/r23_object_message_bler.csv, results/sensitivity/r23_fragmentation_harq.csv, results/baselines/two_gate.csv, results/baselines/two_gate_dgate.csv, results/baselines/importance_map_jscc/two_regime_edge_clean.csv, results/p4b/replay_summary.csv, results/p4b/perclass_ELF.csv, results/manifests/FROZEN_MANIFEST.json and results/channel/bler_sionna.csv.

This document carries material moved out of the main paper for length. It is not a second set of results: the two appendices below are the prior-protocol arms labelled as such in the main text, and the cue table is the definition list of the 21 committed ego-side cues. Section and label names are preserved so that the main text's pointers resolve against this document.

I. Ego-Side Online Task Cues: full definition list

The selector's task-cue vector $z_t \in \mathbb{R}^{21}$ is defined in full below; the main paper lists five representative cues and points here.

TABLE I
Ego-side online task cues used by the CA-TOSG selector ($d = 21$). All cues are computed locally on the ego vehicle before message selection, removing the causality issue that would arise from depending on the collaborator's state.

Category	Cue	Definition
Agent context	num_cavs ego_num_objects	number of CAVs in scene ego-detected objects
Frame shape	ego_origin_lidar_shape_0/1	raw LiDAR tensor shape
Range stats	pcd_num_points pcd_mean_range pcd_max_range pcd_std_range	total point count mean LiDAR range max LiDAR range range std. dev.
Distance bins	pcd_near_20m pcd_mid_20_50m pcd_far_50_80m pcd_very_far_80m	points within 20 m points 20–50 m points 50–80 m points beyond 80 m
Quadrant cnt.	pcd_front_points pcd_back_points pcd_left_points pcd_right_points	front-half points rear-half points left-half points right-half points
Long-range	pcd_front_far_30m pcd_front_far_50m	front points > 30 m front points > 50 m
Density bins	pcd_density_0_20 pcd_density_20_50 pcd_density_50_80	density 0–20 m density 20–50 m density 50–80 m

II. When Are Perception Cues Necessary? LDPC Cliff versus JSCC Graceful Degradation

Protocol note. A prior-protocol arm, not re-evaluated under the frozen P2 protocol: its selector is trained and evaluated inside the codec-comparison pipeline, sharing the retired engine's BLER lookup and 200-realisation CSI convention. Its numbers may not appear in the same sentence as a frozen-replay number, and no main-text claim rests on them.

Under LDPC + QAM a one-line threshold tracks the learned selector (the main paper); does that equivalence survive a feature codec without a sharp SNR cliff? We replace the feature branch with importance-map JSCC [1] and re-run the per-frame pipeline: per-frame JSCC F1 is re-cached at each SNR (1,000 frames per SNR, AWGN and Rayleigh), the $\{L, \text{Feature-JSCC}\}$ oracle relabelled and the selector retrained—no perception-backbone change.

The mechanism is a codec property. The LDPC + QAM feature F1 cliffs over the 8–12 dB decoding transition (onset 8.0 dB), so SNR is highly informative; the learned JSCC feature F1 is essentially flat at ≈ 0.89 across 0–20 dB, so the estimated SNR says almost nothing about whether a feature message will help on a given frame.

This flips the comparison (Table II). Both codecs are evaluated on the same 2,170 test frames under the identical 200-seed protocol, separating (a) whether the graceful-channel gain exists and resides in the cues—estimated leakage-free in-distribution by 5-fold cross-validation, every frame scored by a selector that never saw it and τ tuned on the training folds only—from (b) whether a deployed, validate-frozen selector realises it, under the headline cross-split protocol.

(a) In-distribution. Under LDPC + QAM the AWGN edge over the best SNR threshold is only +0.002 F1 (95% CI $[+0.001, +0.003]$): small, and—as the feature ablation shows—not attributable to the perception cues alone (the main paper).[1] Under JSCC the best SNR threshold collapses to the Fixed-L operating point—it cannot gate on an uninformative SNR—and the cue-based edge, measured by the in-distribution k-fold estimator, widens to +0.022 F1 ($[+0.020, +0.025]$) under AWGN, +0.018 ($[+0.016, +0.020]$) under Rayleigh, and +0.020 ($[+0.018, +0.022]$) under OFDM on the test split, and to +0.012–+0.015 on the Culver-City domain shift (all CIs exclude zero). In the separate 200-realisation held-out comparison on validate frames, the cue-based selector recovers 56–62% of the clairvoyant oracle's headroom over Fixed L across the three channels. The two quantities are distinct and are reported separately: under AWGN the oracle headroom is 0.0291 F1 and the selector recovers +0.0181 of it, with 0.0275 and +0.0158 under Rayleigh and 0.0281 and +0.0158 under OFDM. In-distribution, then, the graceful-channel decision is content-bound and carried by the ego-side cues—a gain no SNR threshold reaches on these frames, and one that (b) below shows does not survive the cross-split move.

(b) Cross-split deployment. The signal does not transfer: deployed on test the frozen selector over-selects the feature action (JSCC C-request rate 0.14 → 0.42) and falls just below Fixed-L—edge −0.004 (AWGN), −0.008 (Rayleigh), −0.005 (OFDM) on test, ≈0 on Culver-City. The gain is present but not portable; capturing it under cross-domain deployment, e.g. by online or self-supervised adaptation [2], is future work.

Whether task-oriented cues are needed is determined by the feature codec's channel response. Under a cliff codec the decision is channel-bound and SNR suffices; under a graceful codec it becomes content-bound and the ego-side cues—which predict whether object-level detection is already adequate—become the operative signal, in-distribution. Closing the cross-split transfer gap (b) is the remaining challenge.

III. Boundary of the Method: a Second Detection Backbone

Convention. This arm was trained and evaluated on the full-collaborator caches, i.e. before the corrigendum of Change-log P0: it is internally consistent (its own budget, its own N_{cw}, its own references) and its relative conclusion stands, but its absolute F1 and payload figures are on a different accounting basis from the single-collaborator main experiment and are not tabulated beside it.

TABLE II
Learned-selector edge over the best SNR-threshold rule on the same 2,170 test frames (200-seed protocol), reported two ways. In-dist. edge: leakage-free 5-fold cross-validation on the evaluation split (each frame scored by a selector that never saw it, τ tuned on the training folds only). Deployed edge: selector trained on validate and frozen, applied cross-split (the headline protocol). In-distribution the JSCC edge is 9–11× the LDPC-cliff edge—an SNR threshold cannot reach it—while under LDPC the margin is the selector's function form, not a cue gain (cues add no significant F1, the main paper). The deployed JSCC edge is negative: the validate-trained selector does not transfer the graceful-channel gain across the domain shift (Appendix II(b)).

Feature codec	Channel	In-dist. edge	Deployed edge
LDPC + QAM (cliff)	AWGN	+0.002 $[+0.001, +0.003]$	−0.002 $[-0.003, -0.001]$
Importance-map JSCC	AWGN	**+0.022** $[+0.020, +0.025]$	−0.004 $[-0.007, -0.001]$
Importance-map JSCC	Rayleigh	**+0.018** $[+0.016, +0.020]$	−0.008 $[-0.011, -0.006]$
Importance-map JSCC	OFDM	**+0.020** $[+0.018, +0.022]$	−0.005 $[-0.008, -0.003]$

Positioning. This appendix reports a limitation, not a generality result. It is placed here because its conclusion is negative: the selection procedure developed in the main text is in-sample effective on a second backbone but does not transfer to held-out splits under the equal-budget protocol. No claim in the main text depends on it.

Every result above uses one perception backbone. To test whether the selection procedure—not the particular features—is what carries the behaviour, we repeat the whole pipeline on a second detector, SECOND [3], under an equal-budget controlled comparison: the feature message is charged the same per-frame channel budget as the mainline ($B_F \equiv 0.99$ Msym, the same N_{cw} and the same BLER tables), so the two backbones are compared at equal channel cost. We do not claim that SECOND's feature tensor compresses to that size, and we declare no codec for it.[2]

Both SECOND checkpoints are the official OpenCOOD ones, and we verify them before use rather than trusting the file: after a lossless axis-order conversion required by the spconv version in our environment, official inference reproduces the published AP@0.7 to +0.0002 on both splits for the late-fusion model (0.7752 vs. 0.775; 0.6822 vs. 0.682) and to within +0.0019 for the intermediate model. The three per-frame branches (E, L, F) are then re-derived with the same scorer, the same canonical union ground truth and the same IoU-0.5 convention as the mainline, and the entire downstream pipeline—grid expansion, scene-level 9-fold LOSO, the budget walk, and the 200-realisation paired replay—is re-run with the mainline code, each stage first verified to reproduce its own committed mainline product bit-for-bit.

The result is that CA-TOSG is in-sample effective but does not transfer under the equal-budget protocol. Table III gives the paired comparison against an SNR-

[1] This in-distribution per-frame edge (+0.002) is distinct from two other LDPC-regime quantities: the feature-ablation gain of the cues (Full vs. channel-state-only, +0.0090, 95% CI $[+0.0089, +0.0091]$, the main paper) and the matched-payload margin over the retuned threshold (+0.0032, the main paper). The three measure the selector along the cue, the equal-bandwidth, and the per-frame axis respectively. Under the corrected single-collaborator convention all three are positive, with the ordering $+0.002 < +0.0032 < +0.0090$: the cue axis, smallest and indistinguishable from zero under the retired full-collaborator accounting (−0.0002), is now the largest of the three — weakening the single message widened the gap the cues have to close, and they close part of it.

[2] The transmitted tensor was measured, not assumed: SECOND's cross-link BEV tensor is 6,758,400 elements per collaborator before compression and 352,000 at the auto-encoder bottleneck. Both are recorded as measurements; neither is used as the operative payload under the equal-budget convention.

TABLE III
Second backbone (SECOND), equal-budget controlled. Paired ΔF1 of the frozen selector against the SNR-threshold rule tuned to the same budget, over the same 200 CSI samplings; 95% frame-level paired bootstrap intervals. Positive favours CA-TOSG. Second-backbone arm, not deployed; descriptive, no decision is taken from this table.

Split	$B_{\max} = 0.10$	$B_{\max} = 0.20$	$B_{\max} = 0.30$
Validate (in-sample)	+0.0058	+0.0095	+0.0081
	[+0.0057, +0.0059]	[+0.0094, +0.0096]	[+0.0081, +0.0082]
OPV2V test	−0.0014	−0.0047	−0.0066
	[−0.0015, −0.0013]	[−0.0048, −0.0046]	[−0.0067, −0.0065]
Culver-City	−0.0029	−0.0117	−0.0167
	[−0.0030, −0.0028]	[−0.0119, −0.0115]	[−0.0169, −0.0165]

threshold rule tuned on validate for the same target budget. On validate—the split the selector is trained on—it is ahead at every budget (+0.0058 to +0.0095 F1). On both held-out splits it is behind at every budget, with all confidence intervals excluding zero.

Two mechanisms account for the gap, and both are visible in the arm's own numbers. First, the L–F margin is narrower on this backbone: SECOND's object-level branch scores 0.869 / 0.879 / 0.879 mean per-frame F1 on validate / test / Culver-City against a compressed-feature branch at 0.897 / 0.904 / 0.939, so there is simply less to gain from choosing correctly, and the same selection error costs proportionally more. Second, the budget walk selects a different candidate family: all three frozen SECOND selectors carry class_weight=balanced, whereas all three mainline selectors carry class_weight=None. The reweighted family fits the in-sample class mix well—agreement with the oracle is 0.833 on validate—but transfers poorly, falling to 0.578 on test and 0.534 on Culver-City, where the selector degenerates to near-always-L (E recall 0.027 on test and 0.000 on Culver-City). The E-collapse of Section IV-G is therefore worse on this backbone, not milder: under Rayleigh the selector's ρ_E is 0.002 (test) and 0.000 (Culver-City) while the oracle spends 0.157 and 0.133. The large payload reductions this arm shows (67–90%) follow from that collapse to L and are not evidence of good selection.

We report this as a boundary of the present method rather than a tuning target: nothing was retrained and no threshold was adjusted after seeing these numbers.

IV. Material moved from the main paper

The four subsections below are reproduced verbatim from the main paper's results section; the main text keeps a short summary of each and points here.

A. Collaboration Is Not Always Beneficial

Collaboration is not unconditionally beneficial, and that is what makes an explicit per-frame selector necessary rather than a convenience. Two failure modes call for two responses. When the channel cannot carry a feature message, requesting one spends the collaborator's transmission budget for nothing and collapses the output to the ego-only floor (the ego vehicle's own pre-fusion detection); the design answer is already in place – the oracle removes any action whose frame-level failure probability is ≥ 0.999 from its feasible set (the same 0.999 constant as the feasibility mask of the main paper), and on failure the pipeline reverts to the ego-only output rather than a phantom feature. When the channel can carry the message, requesting it can still cost accuracy: on the easy stratum the selector's realised output falls below even the always-object-level (Fixed-L) baseline.[3] This mode has no masking answer; its remedy is left to future work. Two CSV-verified quantifiers bound where the ego-side harm sits: the ego-only output strictly exceeds the object-level fused output on 1.5 / 5.8 / 0.2% of validation / test / Culver-City frames, and – from the same per-frame (comp − ego) identity, re-derived on the corrected single-collaborator caches – the compressed-feature message, when delivered, yields lower frame F1 than the ego-only fallback on 1.3 / 6.5 / 0.9% of frames. Test carries the harm most, consistent with fusion having the least to add in thin scenes (mean 15.2 ground-truth objects on test vs 27.8 on validate and 41.0 on Culver-City). No new signalling is needed to act on this: E (ego-only, no cooperative perception payload) is already a deployed action of $\mathcal{S}$ with its own codepoint, so the remedy is a selection-policy question—the frozen selectors almost never choose it (Section IV-G)—not a message-format one.

B. Budget-Matched External Comparison (Where2comm)

What was run, and under which conventions. This is a budget-aligned perception diagnostic under ideal delivery: Where2comm is compared against CA-TOSG on the same frames, the same ground truth and the same scoring chain, at payloads aligned to the same budgets, but the channel model is not applied to either arm's detections here—every message is treated as delivered. The transport-aware comparison is a separate step and is not reported below. Everything that could make the comparison unfair is disclosed rather than buried. The collaborator convention is the mainline one: the nearest single collaborator, applied at inference through the same hook every mainline arm uses, so the comparator is not retrained—retraining it at $N = 1$ would have made it the only arm trained under a different discipline. Its checkpoint is our own 50-epoch reproduction. Scoring uses the frozen chain and a ground-truth set made identical to the mainline's by construction: the two in-dataset filters disagree near the $|y|$ boundary, so ground-truth boxes are matched one-to-one between the

[3]Test Easy stratum (top tercile of the ego's own object-level F1, the §IV-H stratification), evaluated under a deterministic reliable-channel condition (AWGN 16 dB; frame-level BLER ≈0, well above the 8.0 dB onset), isolating the difficulty axis from channel variability: the frozen selector's realised F1 at $B_{\max} = 0.20$ is 0.9749 vs the Fixed-L baseline 0.9796 – a gain of −0.0047 (frame-level paired 95% CI [−0.0074, −0.0024]; $n = 713$ frames). The selector requests F on 241 of these 713 frames; on the remaining 472, where it requests L, its output is frame-identical to Fixed-L, so the paired difference arises entirely on the F-request frames – the loss is a structural consequence of requesting features on already-easy frames.

TABLE IV
Measured mean transmitted fraction against the communication threshold. Between 0.013 and 0.015 the fraction falls by ≈0.23 in one step, and the fraction required by $B_{\max} = 0.20$ (0.1978) lies inside that step on every split. Source: results/baselines/where2comm_v2/sparsity_payload.csv.

threshold	0.01	0.011	0.012	0.013	0.015	0.02	0.05
validate	0.4839	0.4349	0.3875	0.3209	0.1204	0.0831	0.0319
test	0.4648	0.4222	0.3786	0.3130	0.0802	0.0521	0.0179
Culver-City	0.4851	0.4331	0.3825	0.3117	0.1264	0.0908	0.0379

TABLE V
Budget matching. A point counts as matched when its mean payload is within ±20% of the cap. The middle budget is matched on no split; its bracket is the pair of reachable fractions either side of the requirement. Source: results/baselines/where2comm_v2/budget_match.csv.

Split	$B_{\max}$	fraction needed	nearest reachable	error	matched
validate	0.10	0.0967	0.0831	−14.1%	yes
validate	0.20	0.1978	0.1204	−39.1%	no
validate	0.30	0.2988	0.3209	+7.4%	yes
test	0.10	0.0967	0.0802	−17.1%	yes
test	0.20	0.1978	0.3130	+58.3%	no
test	0.30	0.2988	0.3130	+4.8%	yes
Culver-City	0.10	0.0967	0.0908	−6.1%	yes
Culver-City	0.20	0.1978	0.1264	−36.1%	no
Culver-City	0.30	0.2988	0.3117	+4.3%	yes

two canonical sets by box centre (tolerance 0.5 m; every matched pair is at distance 0.000 m, and the counts at 0.01 m are identical), and only matched objects are scored, inside the three-way volume $|x| \leq 70.4$, $|y| \leq 38.4$ m. Payload uses the pre-registered sparse convention: selected feature elements at the declared bit-per-element rate, the index cost charged, and the confidence map not charged—the cheaper reading, which favours the comparator.

The control parameter does not rest where the middle budget is. Where2comm's sparsity is not a setting but an outcome: a confidence threshold is the control, and the transmitted fraction is measured after the fact. Twelve thresholds were run, three of them placed by two refinement rounds inside the interval where the budgets live.

Budget matching, and the cell that could not be matched. A grid point counts as matched when its mean payload is within ±20% of the cap. $B_{\max} = 0.10$ is matched on all three splits (−14.1%, −17.1%, −6.1%; the comparator spends under its cap) and $B_{\max} = 0.30$ on all three (+7.4%, +4.8%, +4.3%; over its cap, so part of any advantage there is bought). $B_{\max} = 0.20$ is matched on none: on test the reachable fractions bracket the required 0.1978 as $[0.0802, 0.3130]$, and the interval is empty.

That cell is the pre-registered confirmatory comparison, so the confirmatory result of this arm is that the budget-matched comparison at $B_{\max} = 0.20$ could not be performed: Where2comm's threshold has no setting placing its payload within ±20% of that cap, and twelve grid points— three of them placed inside the bracket deliberately—leave the interval empty. It is reported as a

TABLE VI
Perception-layer diagnostic, ideal delivery. Comparison at the two matched budgets, AP@0.5, on the intersection ground-truth set inside $|x| \leq 70.4$, $|y| \leq 38.4$ m, with no channel model applied to either arm. This table isolates what the two selections do to detection quality when every message arrives; Table VII adds the transport and reverses every sign. No interval, no margin and no decision is attached to any cell here. Source: results/diagnostics/intersection_gt_track.csv.

$B_{\max}$	Split	Where2comm	CA-TOSG	Δ	Fixed L	fraction
0.10	validate	0.91519	0.90883	+0.00636	0.90934	0.0831
	test	0.94358	0.94490	−0.00132	0.94506	0.0802
	Culver-City	0.89572	0.89508	+0.00064	0.89507	0.0908
0.30	validate	0.91653	0.90930	+0.00723	0.90934	0.3209
	test	0.94417	0.94363	+0.00054	0.94506	0.3130
	Culver-City	0.89885	0.89864	+0.00021	0.89507	0.3117

TABLE VII
Under the modelled transport. The same grid points and the same ground truth, now replayed over the frozen 200 paired CSI draws: same seed and draw order as the mainline, frame BLER re-derived at this arm's own codeword count ($N_{\text{cw}} = 335$–1288 against the mainline's 3960), one shared delivery coin, whole-frame all-or-nothing, and a delivery failure falling back to Where2comm's own ego-only forward. Intervals are paired bootstrap (10,000 resamples, seed 12345, percentile) over the CSI realisations. Source: results/diagnostics/transport_replay_ci.csv.

$B_{\max}$	Split	delivered	Where2comm	CA-TOSG	Δ (95% CI)
0.10	validate	0.3095	0.82529	0.90880	−0.08351 [−0.08374, -
	test	0.3095	0.87437	0.94492	−0.07055 [−0.07076, -
	Culver-City	0.3118	0.81679	0.89508	−0.07830 [−0.07864, -
0.30	validate	0.3055	0.82507	0.90910	−0.08403 [−0.08426, -
	test	0.3056	0.87414	0.94332	−0.06918 [−0.06941, -
	Culver-City	0.3081	0.81729	0.89878	−0.08149 [−0.08182, -

bracketing pair, with no adjudication. This is a statement about the comparator's control parameter, not about either method's perception quality.

How Table VI is to be read. Four points, none of which is optional. First, these cells are descriptive: no interval, no pre-registered margin and no decision is attached to any of them, and the confirmatory cell is the one that could not be run. Second, five of the six cells favour Where2comm by between +0.00021 and +0.00723 AP@0.5 and one favours CA-TOSG by −0.00132; a sign pattern at that magnitude, without intervals, is not a result and is not offered as one. Third, the budget errors run in opposite directions at the two budgets—under the cap at 0.10, over it by 4.3–7.4% at 0.30—so the 0.30 rows are partly bought. Fourth, on test both arms sit at or below Fixed L, which on that split is above the perfect-channel feature ceiling; that is the field-of-view effect of Section IX, reproduced here on a ground-truth set built by a different construction.

Why every sign reverses. Under ideal delivery five of the six cells favour Where2comm by between +0.00021 and +0.00723; under the transport all six favour CA-TOSG by 0.069 to 0.084, an order of magnitude larger and in the opposite direction. The mechanism is in the delivered column. Where2comm transmits on every frame—its sparsity control sets how much to send, never whether—and it has no cheap rung to fall back to, so on the ≈69% of frames the link cannot carry it loses the message entirely and reverts to its own ego-only output. CA-TOSG at these budgets requests the feature branch on a minority of frames and holds the compact object-level message otherwise, so the transport barely moves it. The comparison therefore measures exactly the property this paper argues for: on a cliff channel, choosing whether to send is worth more than choosing what to send within a fixed granularity.

Confirmatory cell, under amendment A2. The pre-registered confirmatory cell (test, $B_{\max} = 0.20$) is unreachable by any single threshold, so it is reached under a registered amendment: a per-frame mixture policy, $p = 0.5052$ of threshold 0.013 with 0.015, whose mean transmitted fraction is 0.1978—the cap exactly—with each frame charged at its own codeword count. Under that amendment, and under the evaluated retransmission and all-or-nothing delivery settings, CA-TOSG attains AP@0.5 0.94492 against Where2comm's 0.87427, a difference of -0.07064 with a 95% paired-bootstrap interval of $[-0.07087, -0.07042]$: an advantage more than an order of magnitude beyond the pre-registered $\delta = 0.005$ margin, attributable to transmission behaviour under this transport rather than to detection quality, which the ideal-delivery table shows running the other way. Three qualifications belong with that sentence and are not optional. The metric is AP@0.5, not the frame F1 the margin was pre-registered on. The interval quantifies stability over channel realisations, conditional on the fixed evaluation set; it does not cover scene-sampling variability and is not the same construction as the frame-level paired-bootstrap interval the main paper's confirmatory comparison uses. And the comparator is a reproduction-grade checkpoint—our own 50-epoch training, not the authors' released weights—applied without retraining.

Orthogonality is unchanged by the comparison. Where2comm decides which feature regions to send within a fixed granularity; CA-TOSG decides whether feature-level information is sent at all. The two compose: a spatial-selection codec can sit inside the feature branch while the granularity selector decides when that branch is worth its payload and channel risk.

C. Collaborator Scale

The evaluation so far treats the collaborator set as given. To ask how the policy behaves as more vehicles are reachable, we sweep the number of requested collaborators $N \in \{1, 2, 3\}$, keeping the frozen selector, $\lambda^\star$ and $\tau^\star$ untouched.

N is not the operative quantity: what matters is the effective count k_{eff}, since requesting three cannot conjure a third. Of the 1,980 validate frames, 733 are saturated at $N = 1$, 1,081 at $N = 2$ and 1,234 at $N = 3$; on Culver-City the supply runs out, so $N = 2$ and $N = 3$ are the same arm. Comparisons are conditioned on the same frames.

F1 then rises with k_{eff} with sharply diminishing returns. At $B_{\max} = 0.20$ on validate, $N = 1 \to 2$ buys +0.0300 F1 for +0.1183 Msym and the third adds +0.0061 for +0.0991 Msym—a fifth of the gain for three-quarters of the channel; on test the pattern is flatter (+0.0096 then +0.0008). The payload sub-linearity is a supply effect: the action distribution is unchanged across N, so the extra channel use comes only from frames that gained a collaborator.

Because "request N" is ambiguous when fewer than N collaborators exist, we bracket the two delivery semantics—charging only for what is delivered, versus charging for what was requested—on validate at $N = 2$, on the 964 frames where the two differ. The bracket is tight: charging for what is delivered rather than for what was requested raises realised F1 by +0.00016, +0.00036 and +0.00099 across the three budgets, at identical payload—the request is what is charged under both—so the conclusions here do not depend on which convention is adopted. Finally, the budget does not transfer automatically. The selector is frozen against an average per-frame budget measured with a single collaborator; total communication grows with the number of collaborators actually served, so the original budget guarantee does not carry over unchanged. This is measurable rather than hypothetical: on validate at $B_{\max} = 0.20$, $N = 3$ spends 0.36842 Msym per frame, well above the 0.20 it was frozen to satisfy — and under the corrected single-collaborator accounting even $N = 2$ (at 0.26937) already breaches it. A deployment that serves more collaborators must therefore re-freeze, or budget on k_{eff} rather than per collaborator.

D. Deployment Robustness and Cost

Four practical imperfections—a noisy SNR estimate, channel-state aging, a stale request, and a misread channel type—degrade the selector only gracefully, because an uncertain channel state falls back to L (Table VIII): SNR-estimation noise up to $\sigma \approx 1$ dB costs ≤ 0.0002 F1; a Jakes model at 60 km/h decorrelates the estimate within ∼10 ms, costing -0.0025 before plateauing; a one-frame-stale decision costs at most -0.0109 under our i.i.d.-per-frame assumption (changing 18.2% of decisions). The near-immunity to estimation noise follows from the channel-type-anchored policy of Section IV-F: perturbing continuous $\hat{\gamma}_t$ rarely flips a decision anchored on the discrete channel-type flag, and under Rayleigh the action remains L regardless. Finally, the deployed Random Forest runs in 52.1 ± 5.6 ms per frame (P95 = 58.3 ms) on a single CPU core. Selector-only latency is 52.1 ms; end-to-end system latency is not measured here. The figure is the slowest of the three frozen selectors over 1,000 batch-1 trials each; the deployment claim must hold for every budget, so the worst case is the one quoted.

Misreading the channel type. The selector takes the channel-type indicator c_t as an input, and c_t carries the single largest Gini share of any feature, so its reliability is a deployment question rather than a detail. We flip c_t on a random fraction of frames and replay the frozen policy (P3 sensitivity arm, channel_misclassification.csv). The

TABLE VIII
Deployment robustness (OPV2V test; realised frame F1). Each row perturbs one imperfection while holding the others ideal; losses are bounded because an uncertain channel state defaults to the robust L message. Re-derived under the single-collaborator convention with the frozen selectors at $B_{\max} = 0.20$ (results/sensitivity/robustness_frozen.csv); the CSI-noise and Jakes models are the ones used previously, so only the selector, grid and utilities changed.

Imperfection	Operating point	ΔF1
SNR-estimation noise σ	≤ 1 dB	-0.0002
	2 dB	-0.0006
	5 dB	-0.0021
CSI aging (Jakes, 60 km/h)	stale by $\sim$10 ms	-0.0025 (then plateaus)
Request delay	1-frame-stale decision	-0.0109 (pessimistic i.i.d. bound)

degradation is graceful and roughly linear in the flip rate: on test at $B_{\max} = 0.20$ the arm's own no-flip baseline is 0.89701 realised F1, falling to 0.89620 at a 5% flip rate, 0.89542 at 10% and 0.89384 at 20%, while the payload remains nearly unchanged (0.14036, 0.14033, 0.14032 and 0.14051 Msym). The asymmetry of the two error directions is what keeps the cost small. A frame misread as Rayleigh biases the selector towards the conservative L, which is safe and merely forgoes a gain. A frame misread as AWGN may lead the selector to request F; the block error probability is then evaluated on the true channel, so the message is likely to be lost and the frame falls back to the ego-only output rather than to L. Only the second direction can lose accuracy, and it loses it to the ego-only floor, not to a corrupted feature map. The arm is a sensitivity replay, not a deployed configuration: its no-flip cell is not the mainline replay's.

A fourth imperfection concerns the cheap message rather than the expensive one. The utilities above model the object-level link as reliable ($\mathrm{BLER}_L = 0$), which is an assumption about the low-rate path, not a measurement. Re-weighting the object-level utility analytically as $\mathrm{eff}'_L = \mathrm{late}\,(1 - \mathrm{BLER}_L) + \mathrm{ego}\,\mathrm{BLER}_L$ over $\mathrm{BLER}_L \in \{0.01, 0.05, 0.10\}$, with every policy frozen and blind to it, leaves the payload — and therefore the payload advantage — untouched by construction, and costs F1 roughly linearly: on test at $B_{\max} = 0.20$, $\mathrm{BLER}_L = 0.10$ costs the selector 0.00634 F1, the threshold rule 0.00588 and Fixed L 0.00734 (results/sensitivity/r23_object_message_bler.csv). The selector's margin over Fixed L therefore widens as the cheap link degrades, while its gap to the threshold rule widens against it, from -0.00010 to -0.00056 — the threshold rule leans harder on the feature action ($\rho_F = 0.20$ against 0.12) and is correspondingly less exposed to an unreliable L. We report this rather than the favourable half alone; it remains inside the pre-registered margin of 0.005 at every BLER_L evaluated.

E. Selector Training

The selector is a Random Forest classifier [4] over the 23-dimensional input $x_t = [z_t, \hat{\gamma}_t, c_t]$, with labels the channel-aware oracle $s^\star_t$ of the oracle definition of the main paper. Selection is budget-indexed and pre-registered rather than hand-tuned: a fixed block of 112 candidates (hyperparameter settings × a payload-penalty grid λ) is evaluated once under scene-level 9-fold leave-one-scene-out cross-validation on validate—each of the nine scenes held out exactly once, 1,008 fold rows—and for each budget $B_{\max} \in \{0.10, 0.20, 0.30\}$ the candidate list is walked in descending out-of-fold F1 until one meets the budget, which freezes that budget's $(\lambda^\star, \tau^\star)$ and hyperparameters. The three frozen selectors are $\lambda^\star = 0.05/0.02/0.00$ with $\tau^\star = 18/12/8$ dB at $B_{\max} = 0.10/0.20/0.30$; all three carry class_weight=None—the balanced candidates were walked past for exceeding their budget—and FROZEN_MANIFEST.json is the authority for every one of these values. Each frozen configuration is then refitted once on the full 1,980-frame validate split and frozen—never retrained—for every reported test and Culver-City result. Run-to-run variance is exposed not by a within-validate resplit but by averaging all realised-F1 and payload metrics over 200 Monte-Carlo channel realisations per operating point (per-frame SNR $\sim \mathcal{U}[0, 20]$ dB and channel type $\sim$ Bernoulli(0.5) AWGN/Rayleigh).

F. Selector Decisions and Feature Importance

Fig. 1 shows the per-mode selection ratios ρ_s against SNR, with the frozen selector overlaid on that budget's payload-penalised oracle and the stacked-area panel showing the share of each action of $\mathcal{S} = \{E, L, F\}$ at every SNR. Under AWGN the selector's adaptation knee is at 10 dB, where the frame-error cliff clears: ρ_F steps from 0 to 0.472 on validate (0.433 on test, 0.160 on Culver-City) and is flat above it, with ρ_L falling correspondingly. Under Rayleigh ρ_L stays at essentially 1 across the whole range. The panel also makes the E-collapse of Section IV-G visible: under Rayleigh the oracle spends a substantial share on the ego-only action—$\rho_E = 0.157$ on test and 0.133 on Culver-City—while the frozen selector's ρ_E is 0.002 and 0.000. E is a deployed action that the imitation objective almost never selects, which is a limitation of the trained policy rather than of the action set. It is, however, not a limitation a simpler policy repairs: a hand rule with an explicit E gate, fitted over three difficulty proxies in both orientations, selects E in no split × budget cell ($\rho_E = 0.00000$ throughout, the main paper) — on these cues E is structurally out of reach of a monotone threshold, and the frozen selector's small but non-zero ρ_E is more than any of them attains. Under the corrected single-collaborator convention this limitation became cheaper without becoming milder. The behaviour is unchanged — the selector still almost never chooses E — but the price of those missed calls fell from 0.58–0.61δ to 0.44–0.47δ on test, where $\delta = 0.005$ is the pre-registered non-inferiority margin. The oracle's own appetite for E moved in opposite directions on the two splits: the number of grid cells it labels E rose on validate (335 → 529) and fell on test (6,303 → 5,696), so the correction does not simply make ego-only more attractive everywhere.

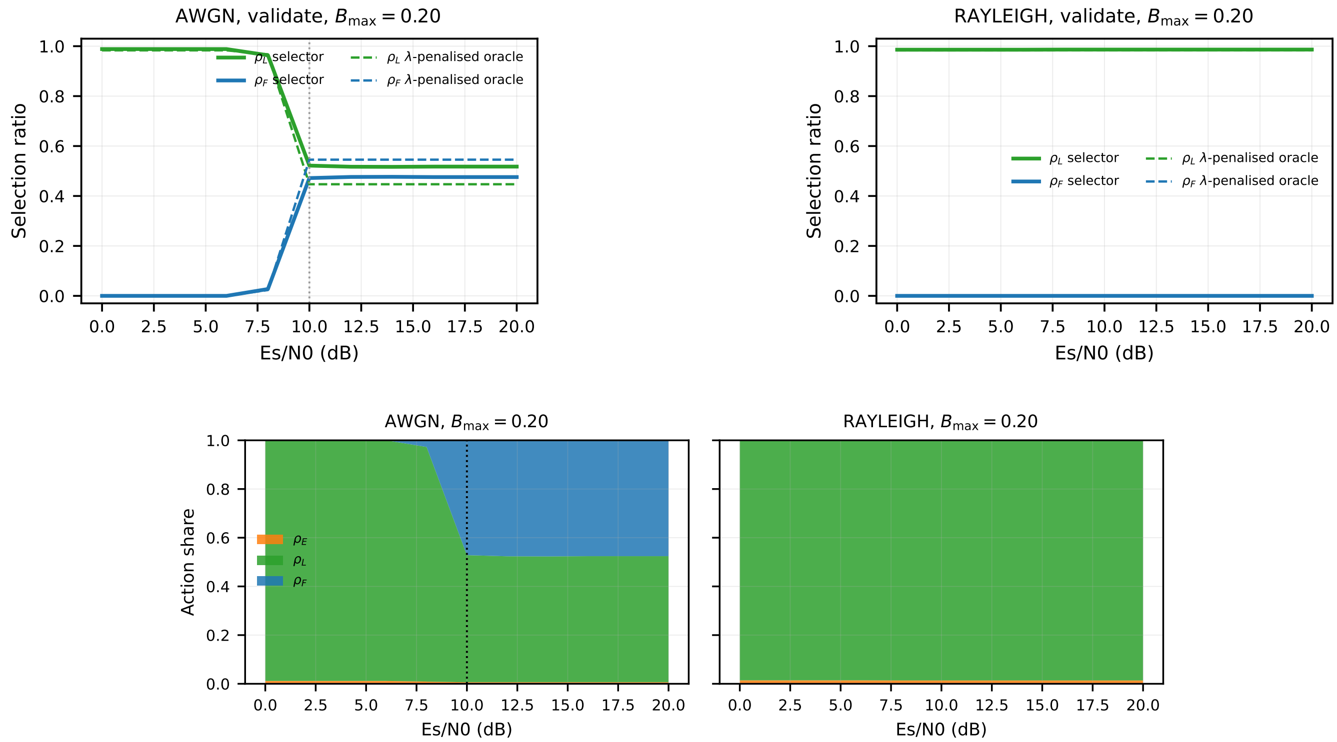

Fig. 1. Per-mode selection ratios versus SNR on OPV2V validate at $B_{\max} = 0.20$, from the frozen grid. Top: the frozen selector (solid) against that budget's λ-penalised oracle (dashed) — the oracle sees the frame BLER, not the per-frame block outcome, so it is not clairvoyant. Bottom: stacked-area view of the selector's action distribution over the deployed action set $\mathcal{S} = \{E, L, F\}$ under AWGN and Rayleigh (the non-deployed C_{256} physical-layer comparator); the dotted line marks the measured 10 dB policy knee. Two readings: under AWGN ρ_F steps from 0 to 0.472 at 10 dB and is flat above it, and under Rayleigh the selector's ρ_E is 0.002 (test) / 0.000 (Culver-City) while the oracle spends $\rho_E = 0.157$ / 0.133 there — the E-collapse limitation of Section IV-G. The three-budget version is the per-budget action distributions (repository figure product).

Feature importance.

the feature-importance table of the main paper report the top Gini feature importances of the deployed selector. The two channel-side features dominate: the channel-type indicator c_t contributes 34.2% of importance and the estimated SNR $\hat{\gamma}_t$ a further 27.5%, so two of the 23 inputs carry 61.7% between them against 38.3% spread over the 21 perception-side cues. The strongest of those, pcd_mean_range, reaches only 3.0%. Two cautions belong with that number. Importance is not sufficiency: a Gini share says how often a feature is split on, not that the remaining 38.3% is redundant, and the feature ablation shows the opposite — with the channel features alone the selector stops requesting features at the tighter budgets and never recovers the full selector's operating point (the main paper). And the split is transport-specific: it is measured under an LDPC + QAM cliff where the channel state is nearly decisive by construction; under graceful degradation the same cues carry the decision instead (the supplementary material). That the channel-type indicator c_t outranks the estimated SNR $\hat{\gamma}_t$ is consistent with the frame-level channel physics: under Rayleigh fading the feasible set collapses to L regardless of SNR, so the channel-type flag is the single highest-information split for the decision.

G. Communication–Perception Pareto Frontier

The headline table compares policies at a single operating point. To show that CA-TOSG sits on the attainable payload–F1 frontier rather than merely better on average, we sweep the Lagrangian selector $s_t^\star = \arg\max_s (F_t^s - \lambda B_s)$ over λ and plot every policy on the payload–F1 plane (Fig. 2). As λ grows from 0 (max F1) to large values (min payload), the oracle traces the achievable upper frontier; the fixed feature-level policies C_{16} and C_{256} sit far below it (high payload, low realised F1 due to the LDPC cliff), i.e. they are strictly dominated. CA-TOSG and a λ-cost RF retrained on $\arg\max_s (F_t^s - \lambda B_s)$ labels lie on the frontier between Fixed L and the oracle. The P2 product is three frozen selectors, one per budget, so the operating point is budget-indexed: on test they realise F1 0.89148, 0.89691 and 0.89783 at 0.037, 0.141 and 0.21196 Msym/frame for $B_{\max} = 0.10$, 0.20 and 0.30. At $B_{\max} = 0.20$ that payload is 14.3% of the feature-level payload B_F.

H. Where the Gain Concentrates: Difficulty Stratification

A channel-averaged mean masks where CA-TOSG matters. We stratify frames into easy/medium/hard terciles by the ego's own object-level detection F1 (low = hard) and evaluate the frozen selectors at a single reliable-channel operating point—deterministic AWGN at 16 dB,

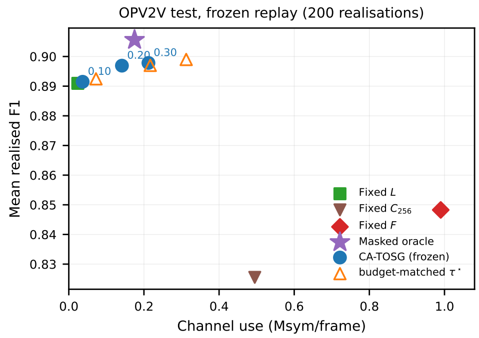

Fig. 2. Communication–perception plane on OPV2V test, from the frozen 200-realisation replay. The axis spans the full range so Fixed F (0.99 Msym) is visible rather than clipped. The three frozen selectors are shown budget-indexed (circles, labelled by $B_{\max}$) beside the SNR-threshold rule tuned to the same budget (open triangles); the feasibility-masked oracle is a separate marker (star). Fixed F and Fixed C_{256} are dominated by Fixed L at far higher channel use.

one point of the pre-registered SNR grid rather than a channel draw (the difficulty stratification (repository figure product)). At $B_{\max} = 0.20$ the CA-TOSG-over-Fixed-L gain rises monotonically with difficulty: on test it is -0.0047 (95% CI $[-0.0074, -0.0024]$) on easy frames, $+0.0056$ ($[+0.0027, +0.0085]$) on medium frames and $+\mathbf{0.0660}$ ($[+0.0591, +0.0730]$) on hard frames. Validate has the same shape ($+0.0058/+0.0441/+0.0470$); Culver-City is weaker throughout ($+0.0011/+0.0021/+0.0310$). The value of CA-TOSG is therefore two-dimensional: it materialises where a hard frame meets a reliable channel, and a channel-averaged headline necessarily dilutes it. On easy frames the deployed selector over-requests F on test, and the frozen product puts the cost at -0.00471 F1 (95% CI $[-0.00742, -0.00236]$) at $B_{\max} = 0.20$, growing monotonically with the budget —-0.00020, -0.00471, -0.01129 across 0.10/0.20/0.30—as the looser cap lets more easy frames be sent as F. The effect is split-dependent rather than universal: on validate and Culver-City the same stratum gains at every budget ($+0.00035$ to $+0.01016$ and 0.00000 to $+0.00558$), so it is a property of the test split's easy tercile, where fixed L already reaches 0.97962 and there is almost nothing left to buy. A payload-penalised ($\lambda > 0$) operating point on the Lagrangian frontier of Section IV-G would mitigate it.[4]

V. Feature-ablation comparison table

The table the ablation section of the main paper points to.

VI. Tables moved from the main paper

The two tables the main paper points to; its prose carries their conclusions and numbers.

[4]The channel-averaged companion view reported in an earlier version of this subsection is withdrawn rather than recomputed: it was a product of the retired 200-realisation policy engine and its selector, not of the frozen selectors evaluated here, and the two are different quantities that may not be placed side by side.

TABLE IX
Selector feature ablation and the SNR-threshold challenger (train: validate, eval: frozen OPV2V test at $B_{\max} = 0.20$, $T = 2{,}170$ frames, 200 SNR/channel realisations). Each variant is retrained through the identical selection pipeline—same candidate set, same scene-level 9-fold LOSO, same frozen walk under the budget—with only the feature columns changed. Neither half of the input alone produces a graded policy: both the channel-only and the cues-only variants collapse to always transmitting L at this budget (feature-request rate $\rho_F = 0$), so only the full 23-feature selector reaches an intermediate operating point. The cues-plus-$\hat{\gamma}$ cut of the earlier ablation was not re-run under the corrected protocol and is therefore not listed.

Selector	#feat	Realised F1	Deployed payload (Msym)
Channel only ($\hat{\gamma}, c$)	2	0.89095	0.02400
Perception cues only	21	0.89092	0.02392
Full (all features)	23	0.89691	0.14141
SNR-threshold ($\tau^\star$)	–	0.89701	0.21679
Fixed L	–	0.89095	0.02400

TABLE X
Notation for the semantic granularity levels. F and C_{16} denote the same message; the C_q form is used only where the modulation order is the subject.

Symbol	Meaning	Channel use	Role
E	ego-only, no transmission	$B_E = 0$	deployed action; also the deli
L	object-level message	$B_L = 0.024$	deployed action
$F \equiv C_{16}$	feature-level, 16-QAM	$B_F = 0.99$	deployed action
C_{256}	feature-level, 256-QAM	0.495	physical-layer comparator, nc

VII. True end-to-end AP verification: full protocol and readings

The verification prose the main paper compresses to two sentences.

A. True End-to-end AP Verification

The realised frame F1 reported so far is computed under the analytical block-loss model of Eq. (4) of the main paper, where a lost feature block falls back to the ego vehicle's own pre-fusion detection. To confirm that this analytical F1 proxy translates to actual end-to-end detection AP under a deployed pipeline, we run a true end-to-end inference in which: (i) the deployed selector picks s_t per frame from $(\hat{\gamma}_t, c_t)$; (ii) cached per-frame predictions from the OPV2V late-fusion checkpoint (for L) and the attentive-compression checkpoint (for C_q) are concatenated according to the selector's choice; (iii) each C_q frame is dropped with probability $\mathrm{BLER}_q(\gamma_t, \mathrm{ch}_t)$ and falls back to the ego-only (pre-fusion) prediction; (iv) the resulting box set is scored against ground-truth boxes with the OpenCOOD VOC-style AP computation at IoU thresholds $\{0.3, 0.5, 0.7\}$, averaged over the 200 paired stochastic channel realisations of the frozen replay. The per-SNR breakdown is the committed product true_e2e_ap_by_snr.csv.

Two observations support the central claim. First, the boundary of the feature-active regime is measured, not stipulated: reading the frozen selector's feature-request rate off the pre-registered SNR grid, ρ_F is zero at every

TABLE XI
Pre-registered sensitivity analyses (test at $B_{\max} = 0.20$ unless stated). Products: r23_scene_bootstrap.csv, r23_object_message_bler.csv, r23_fragmentation_harq.csv, r25_fragmentation_replay.csv.

Check	Quantity	Result
Scene-level bootstrap	ΔF1 95% CI	$[-0.001508, +0.001522]$ (18.93× wider)
	ΔB 95% CI	$[-0.119307, -0.029856]$ (45.177×)
BLER$_L = 0.10$	F1 cost, selector	-0.00634 (threshold -0.00588, Fixed L -0.00734)
	payload	unchanged by construction
Fragmentation + HARQ	AWGN 8 dB BLER	$0.402400 \to 0.100363$ ($k=2$), 0.050777 ($k=4$)
	payload factor	1.226954×, 1.120770×
	Rayleigh 0–20 dB	frame BLER 1.0 for every k

AWGN point up to 8 dB and steps to its plateau at 10 dB on all three splits—0.472 on validate, 0.433 on test and 0.160 on Culver-City—which is exactly where the 16-QAM frame-error cliff clears (Section III-C of the main paper).[5] At that boundary the true end-to-end AP on validate is 0.8287 at IoU 0.5 and 0.7292 at IoU 0.7, and it is flat above it. The selector reaches this while paying 0.300 Msym/frame averaged over the feature-active regime instead of the 0.990 of always-on Fixed F—a 70% saving—though on validate it recovers only part of the perfect-channel reference of the underlying attentive-compression model (0.8369 on validate, Section VI-A of the main paper) rather than essentially all of it, as an earlier version of this section claimed. Second, the Rayleigh AP curve is flat across the full SNR range and matches Fixed L, confirming that the selector's conservative fading-regime behaviour, identified in Section VI-A of the main paper, is not a frame-F1 artefact but a genuine end-to-end property.

VIII. Figure and table moved from the main paper (R41)

The difficulty/importance figure and the feature-importance table; the main paper carries their conclusions and numbers.

IX. Common-Volume Diagnostic: How Much of the Headroom Is Field of View?

Motivation. The branches of this paper are separate trained pipelines, and one of the four ways they differ is the configured perception range: the late-fusion checkpoint behind L covers $x \in [-70.4, 70.4]$ m, the compression checkpoint behind F covers $x \in [-140.8, 140.8]$ m. Both are scored against one canonical ground-truth set, which extends to $|x| \approx 119$ m. Ground truth beyond the L branch's range is therefore a structural miss for L and a detectable object for F, so part of the measured L–F gap is field of view rather than semantic granularity. The size of that part was unknown, so we measured it.

[5] Validate shows a 0.4% toe at 8 dB, the single grid point at which the frame BLER is neither ≈0 nor ≈1 (0.402); test and Culver-City are exactly zero there. We therefore take 10 dB as the regime boundary on all three splits and report the toe rather than smoothing it away. Earlier versions of this section stipulated ≥ 14 dB; that threshold was not derived from any measurement and is withdrawn.

Method. Every prediction and every ground-truth box, from the same committed caches the headline table uses, is cropped to the intersection volume $|x| \le 70.4$, $|y| \le 40$ m by box centre, and the unchanged scorer is re-run. Two caveats belong with the numbers: the crop is centre-based, so a box straddling the boundary is kept or dropped whole, and the CA-TOSG rows use 20 of the 200 paired realisations, the three fixed references being deterministic. The percentage of canonical ground truth removed is 18.89% on validate, 9.94% on test and 20.48% on Culver-City.

Reading. Every arm's absolute AP rises inside the common volume, which is expected: the removed ground truth is ground truth no branch could reach. What matters is the gap. On validate the headroom falls from 0.0550 to 0.0117, on Culver-City from 0.0970 to 0.0252, and on test it changes sign, from 0.0240 to -0.0061—inside a shared field of view the perfect-channel feature ceiling is below fixed L on test.

Relation to the frozen track. The two tracks share a ground-truth denominator and answer different questions. The deployed track asks what each branch delivers as configured, which is the question a deployment faces and is what every headline number in the main paper reports. The common-volume track asks how much of the L–F gap survives when the field of view is held equal, which is the question a reader asking "is this a granularity effect?" is really asking. Neither supersedes the other, and the diagnostic changes no number in the main paper. It does bound one claim: on test, the headroom a granularity policy could contest is a field-of-view effect.

X. Channel-averaged comparison table moved from the main paper (R42)

The main paper reads this table in two sentences; the full row set is here.

References

[1] Y. Sheng, H. Ye, L. Liang, S. Jin, and G. Y. Li, "Semantic communication for cooperative perception based on importance map," Journal of the Franklin Institute, vol. 361, no. 6, p. 106739, 2024.

[2] C. Liu, J. Chen, Y. Chen, R. Payton, M. Riley, and S.-H. Yang, "Self-supervised adaptive weighting for cooperative perception in V2V communications," IEEE Trans. Intell. Veh., vol. 9, no. 2, pp. 3569–3580, 2024, dOI: 10.1109/TIV.2023.3345035.

[3] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," Sensors, vol. 18, no. 10, p. 3337, 2018.

[4] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.

TABLE XII

Generalisation of the validate-trained frozen selectors (no retraining) to the OPV2V test ($T = 2{,}170$ frames, scene-disjoint) and Culver-City ($T = 550$ frames, real-road-layout domain shift) splits. Every row is the mean over the same 200 paired CSI samplings of the frozen replay (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh); each frozen selector is paired with the SNR-threshold rule tuned to the same budget. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
OPV2V test ($T = 2{,}170$)			
Fixed L	0.8910	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.8255	0.495	50.0%
CA-TOSG $B_{\max}$=0.10 / $\tau^\star$	0.89148 / 0.89247	0.0368 / 0.0724	3.7%/7.3%
CA-TOSG $B_{\max}$=0.20 / $\tau^\star$	0.89691 / 0.89701	0.1414 / 0.2168	14.3%/21.9%
CA-TOSG $B_{\max}$=0.30 / $\tau^\star$	0.89783 / 0.89900	0.2120 / 0.3125	21.4%/31.6%
Channel-aware oracle (masked)	0.9056	0.175	17.7%
OPV2V Culver-City ($T = 550$)			
Fixed L	0.8466	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8099	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.7793	0.495	50.0%
CA-TOSG $B_{\max}$=0.10 / $\tau^\star$	0.84667 / 0.84956	0.0244 / 0.0718	2.5%/7.2%
CA-TOSG $B_{\max}$=0.20 / $\tau^\star$	0.84975 / 0.85858	0.0675 / 0.2174	6.8%/22.0%
CA-TOSG $B_{\max}$=0.30 / $\tau^\star$	0.85932 / 0.86316	0.1823 / 0.3146	18.4%/31.8%
Channel-aware oracle (masked)	0.8643	0.266	26.8%

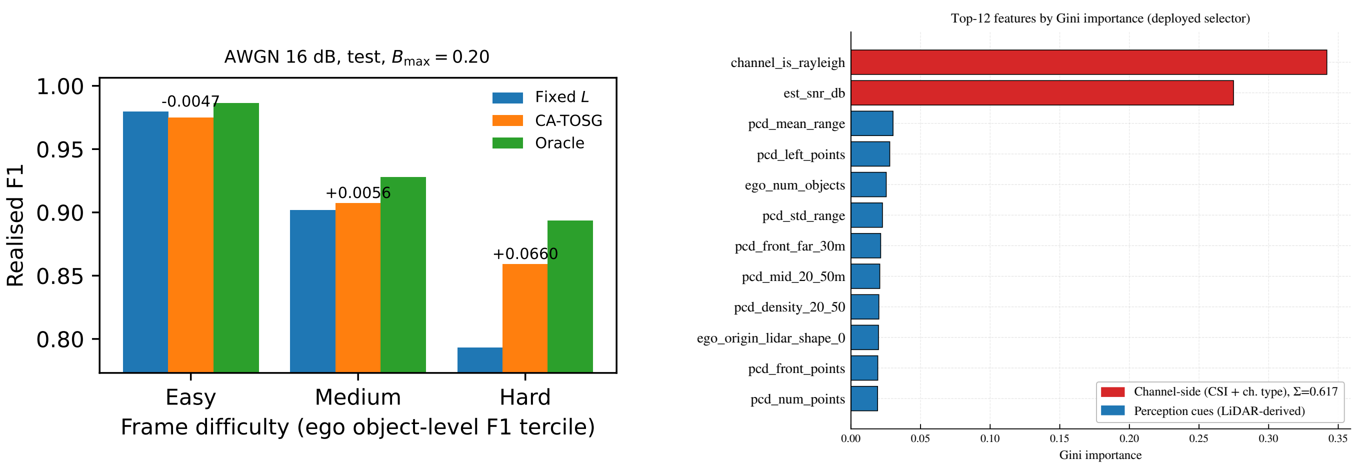

Fig. 3. Left: realised F1 by frame-difficulty tercile (ego object-level F1) on OPV2V test at $B_{\max} = 0.20$ at a single reliable-channel grid point (AWGN 16 dB) with the frozen selector; numbers above bars are CA-TOSG−Fixed-L, the hard-frame gain being +0.0660 (95% CI [+0.0591, +0.0730]). Right: top-12 features by Gini importance of the frozen $B_{\max} = 0.20$ selector—the channel-side features jointly account for 61.7%, the 21 perception-side cues for the remaining 38.3%, none above 3.0% alone. Importance is not sufficiency (the ablation section of the main paper).

TABLE XIII

Top-7 features by Gini importance for the deployed CA-TOSG selector (trained on OPV2V validate, $T = 1{,}980$ frames). The two channel-side features account for 61.7% of total importance; the 21 perception-side cues carry the remaining 38.3%. Read directly from the frozen model, results/main/feature_importance_frozen.csv.

Feature	Gini importance
channel_is_rayleigh	0.342
est_snr_db	0.275
pcd_mean_range	0.030
pcd_left_points	0.028
ego_num_objects	0.025
pcd_std_range	0.023
pcd_front_far_30m	0.021
$\hat{\gamma}_t + c_t$ subtotal	0.617

TABLE XIV
Common-volume diagnostic, AP@0.5. Deployed is the frozen table (each branch as configured); common volume re-scores the same cached outputs inside $|x| \leq 70.4$, $|y| \leq 40$ m. The headroom row is the feature ceiling minus fixed L. Source: results/diagnostics/common_volume_ap.csv.

Split	Policy	Deployed	Common volume	Δ
validate	ego-only	0.6116	0.7539	+0.1423
	Fixed L	0.7819	0.9064	+0.1245
	Feature ceiling	0.8369	0.9181	+0.0812
	headroom	0.0550	0.0117	−0.0433
test	ego-only	0.7350	0.8162	+0.0812
	Fixed L	0.8691	0.9429	+0.0738
	Feature ceiling	0.8931	0.9368	+0.0437
	headroom	0.0240	−0.0061	−0.0301
Culver-City	ego-only	0.6185	0.7775	+0.1590
	Fixed L	0.7299	0.8825	+0.1526
	Feature ceiling	0.8269	0.9077	+0.0808
	headroom	0.0970	0.0252	−0.0718

TABLE XV

Channel-averaged comparison on the frozen OPV2V test split (selector trained on validate, not retrained; $T = 2{,}170$ frames), read from the frozen replay: F1 and channel use are the means over the same 200 paired CSI samplings (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh) for every row. The comparison is budget-indexed: each frozen selector is shown against the SNR-threshold rule tuned to the same budget, and it does not run one way. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
Fixed L (object-level)	0.8910	0.024	2.4%
Fixed F (LDPC+16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC+256-QAM)	0.8255	0.495	50.0%
Masked oracle (upper bound)	0.9056	0.175	17.7%
$B_{\max} = 0.10$			
SNR-threshold ($\tau^\star$=18)	0.89247	0.0724	7.3%
CA-TOSG (RF, frozen)	0.89148	0.0368	**3.7**%
$B_{\max} = 0.20$			
SNR-threshold ($\tau^\star$=12)	0.89701	0.2168	21.9%
CA-TOSG (RF, frozen)	0.89691	0.1414	**14.3**%
$B_{\max} = 0.30$			
SNR-threshold ($\tau^\star$=8)	0.89900	0.3125	31.6%
CA-TOSG (RF, frozen)	0.89783	0.2120	**21.4**%